



APPLICATION OF FUSED FILAMENT FABRICATION IN MARINE SECTOR, FROM RAPID PROTOTYPING TO FINAL PRODUCT

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ABSTRACT

Additive Manufacturing (AM), due to its significant advantages over conventional manufacturing technique has gained much popularity over the years. Especially, due to its ability to manufacture complex geometries with reduced material waste has made AM more appealing to adopt in industrial scale manufacturing. Among the various AM techniques, Fused Filament Fabrication (FFF) due to its simplicity and inexpensive operation and material cost is being explored vastly to use in product manufacturing in different sectors. This paper reviews recent advancements and research in the marine sector through FFF. It also examines the limitations of its use and briefly reviews recent attempts to overcome some of these limitations. Additionally, this paper identifies future research directions that focus on the applicability of FFF in the marine sector.

Keywords: Additive manufacturing, Fused filament fabrication, Prototyping

1. INTRODUCTION

Additive manufacturing (AM), also known as 3D printing, is essentially a layer-by-layer fabrication technique. This technology was first introduced by Charles W. Hull in 1984 through his first working 3D printer. He coined the term “*Stereolithography Apparatus*” [8]. Initially, AM was expensive and thus had limited use and accessibility [30]. However, the development of open-source printers with self-replicating rapid prototypers (known as *RepRaps*) accelerated innovations and competition leading to the technology being affordable to consumers [32]. The various advantages of AM over conventional manufacturing in terms of ease of use, cost-effectiveness, sustainable production, less resource requirement, and design freedom [5, 13, 20, 47] have made it quite appealing in the industries.

Fused Filament Fabrication (FFF) is a low-cost method among AM techniques. The freedom to use open-source technologies, inexpensive raw materials, and parts, along with relatively easier post-processing requirements, have led to various research and applications of FFF in various sectors. In fact, during COVID-19, a significant growth in FFF’s use has been observed, especially in personal protective equipment (PPE) fabrication [46]. Significant applications of FFF have been reported in aerospace [27, 29, 35], automotive [21, 37, 49] and medical industries [10]. Even in the marine sector, FFF has been widely accepted

[15]. From rapid prototyping to patching and repairs, researchers are exploring FFF’s potential in this sector even in fabricating end parts.

In this paper, the recent accomplishments of FFF in the marine sector are discussed. While FFF’s application in this sector has significantly increased, certain limitations of the process still hinder its further use. This paper reviews those drawbacks and explores the recent attempts in general to overcome these issues.

2. FFF OVERVIEW

In the FFF process, a three-dimensional (3D) model is initially created using Computer Aided Design (CAD) software. This is converted into *Stereolithographic* (STL) file format or *Additive Manufacturing Format* (AMF), which is then exported to a slicing software [18]. The Slicing software, as the name suggests, slices the model into layers, making it printable. The file is then brought to the 3D Printer.

To print the model, material in the form of filament is pressure fed into a heated extruder nozzle, which deposits the molten material on top of a build platform, which may or may not be heated depending on the deposited material. Generally, the extruder moves in the horizontal/XY plane, creating a single layer for the part. Afterwards, the build platform moves in the Vertical/Z direction, enabling successive layer printing. The schematic of the process is shown in Fig. 1.

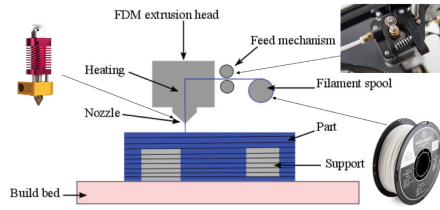


Figure 1. FFF Process Schematic [3]

While this is in general, the operation followed in FFF process, there are several other steps practiced in research and industries, such as pre-processing of the filament and post-processing of the final part to achieve desired properties.

3. FFF IN MARINE SECTOR

3.1 Rapid Prototyping

Prototyping is a crucial step in the process of developing and manufacturing a product. It is necessary to evaluate the shape, compatibility, and performance of a design before investing a substantial amount of money in equipment [48]. Currently, rapid prototyping is the leading method used to create physical models for testing and evaluation [40].

Rapid prototyping (RP) refers to a range of technologies that allow the creation of components without using traditional tools or the necessity for professional model-makers [25]. Broadly, RP technologies can be categorized into two main types: *additive processes*, which involve adding material, and *subtractive processes*, which involve removing material [31]. FFF falls under the additive process classification of RP.

The material properties of 3D printed materials and their ability to duplicate ship structural performance have been extensively evaluated [7, 45]. Printing a rigid model costs more than building one using typical manufacturing methods. However, these shapes allow for complicated structures that can reveal new insights into ship fluid dynamics and structural integrity [7].

Researchers are utilizing the increasing accessibility of cost-effective 3D printing technology to quickly create underwater vehicle parts prototypes for ecological monitoring [33]. For instance, propellers are complex geometries; thus, leveraging additive manufacturing techniques can be beneficial. Considering the precision achieved, desired tensile strength, and printer affordability, FFF is among the suitable choices [14]. Propellers printed with FFF 3D printing were used in oceanography studies nearly a decade ago. A high-drag Autonomous Underwater Vehicle (AUV) used these propellers composed of Acrylonitrile Butadiene Styrene (ABS) and polycarbonate [16]. Fig. 2 demonstrates such a propeller fabricated.

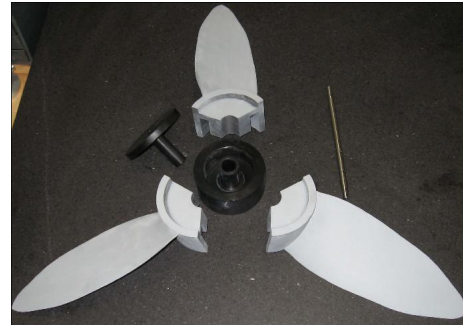


Figure 2. UAV Propeller Printed Using FFF [16]

Other components of such underwater vehicles can also be printed using FFF. For example, the development of a low-drag Stingray AUV for ecological monitoring incorporated the use of 3D-printed plastic components, specifically wings and tail [6].

3.2 Application of FFF in Marine Repairs

AM's capacity to add material makes it a strong contender for repairing parts alongside traditional machining. By adding material to the damaged area, AM allows for further machining to get the desired final shape. Even though FFF can not be used for everything in this case, developing FFF as a repair method will open up interesting options to current thermoplastic repair methods, allowing repairs to be applied automatically and precisely [11]. If the shape of the object permits, a substantial and significant crack can be repaired by applying a patch made of FFF material. Eventually, it is possible to print broken and missing characteristics into the surfaces of the parts [12]. In Fig. 3, stitch patterns are shown for different materials using FFF in repair works.

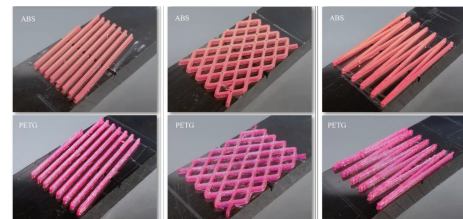


Figure 3. FFF printed patches with ABS (top) and PETG (bottom) in different patterns [12]

Considering the potential of FFF, its possible use in marine repairs, especially onboard ships for spare part manufacturing, has been explored [52]. The earliest attempt of such work dates back to 2016 through the cooperation of Green Ship of the Future [1]. Through this, an endeavor was undertaken to introduce 3D printers onto ships by outfitting six ships and three rigs with desktop 3D printers with FFF technology. The crews have been instructed to evaluate novel technology and manufacture non-essential ship components that are frequently prone to damage or challenging to procure. Some examples of items that have been

printed during this test include a powder extinguisher end nozzle, lathe lever knobs, DIN (Deutsche Institut für Normung) rail fuse mounts, and handrail brackets for the engine control room. The 3D-printed components are shown in Fig. 4

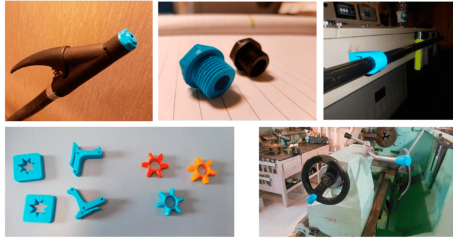


Figure 4. Components fabricated by FFF [1]

However, using FFF in such works requires careful modelling; otherwise, errors in the model can arise, rendering the model unusable. So, when paired with a 3D printing system, the usage of portable 3D scanners offers significant versatility, enabling on-board repairs to be completed quickly. A 3D scanner is used to reverse engineer the parts that require repair and produce a printable design for appropriate maintenance [28].

3.3 Application of FFF in Ship Building

The direct use of FFF in shipbuilding is still very limited. However, FFF has possible applications for small and medium-scale boats, such as yachts, which rely on Fiber Reinforced Polymer (FRP) molding technology [43]. Currently, the yacht business is predominantly semi-custom, meaning that purchasers select vessels based on their size and cost and then subsequently personalize them by choosing interior configurations, design finishes, and equipment packages. While technically feasible, hull alteration is frequently cost-prohibitive for an average purchaser. Provided AM can be scaled up and provided the 3D printed materials fit maritime regulations, it has the potential to enhance yacht hull personalization at a reduced cost [52].

University of Washington (UW) student 3D printing club utilized FFF AM in 2012 to build a small watercraft for a local regatta tournament [24]. The students from the UW Fabbers team conducted research on methods for processing HDPE material obtained from discarded milk jugs. They successfully designed and implemented a customized extruder, which they then installed on an industrial CNC plasma cutter. The UW Fabbers team utilized collected containers to fabricate a straightforward vessel measuring 2.13 meters (7 feet) in length.

The Sacs Marine 700 Series console, an award-winning experiment, utilizes computer modeling and AM to create a marine industry product. The HSM plastic console was designed and built in under four weeks by a small crew, including interns from the

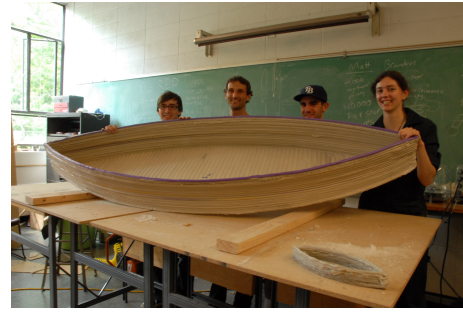


Figure 5. UW team with their printed boat [24]

University of Genoa's Design Navale e Nautico programme. Due to its complicated surface topology, the console cannot be manufactured using typical FRP molding procedures. Superfici's patented componentry technology was used to design and print many subcomponents utilizing FFF on conventional or commercial 3D printing equipment [9].

4. CHALLENGES OF FFF IN MARINE SECTOR

Although FFF exhibits considerable promise in the marine industry, it also faces certain obstacles. A primary problem revolves around the durability of the material when exposed to extreme environmental conditions. Polymer feedstock materials used in FFF for maritime applications must have the necessary resistance to deterioration [34]. Multiple studies have demonstrated a decline in the mechanical characteristics of thermoplastics when exposed to extreme conditions, rendering it challenging to use FFF-printed parts directly. Several studies have demonstrated a decline in the mechanical characteristics of thermoplastics when exposed to extreme conditions, rendering it challenging to use FFF-printed parts directly [4, 22, 34, 36, 44, 50].

Furthermore, components produced using FFF are recognized for their substandard mechanical characteristics [2, 19]. Nevertheless, the implementation of post-processing procedures, such as annealing, results in enhanced mechanical characteristics [39]. However, it should be noted that post-processing is an intensive process that necessitates the use of extra equipment, making it relatively costly [38].

Another concern pertains to the machine's performance in challenging environmental conditions. Base vibration leads to subpar part quality regarding surface roughness and accuracy [26]. In Fig. 6 illustrates the impact of vibration, with the base vibration revealing a coarse surface.

Also, individuals must possess diverse technical and engineering abilities to use 3D printing effectively. This includes being skilled in contemporary design methodologies, having a deep understanding



Figure 6. Surface roughness of FFF printed parts without (top) and with (bottom) base vibration [26]

of novel material processing methods, and being proficient in data management for testing and process monitoring. To ensure that 3D printing has a significant and positive effect on production in the maritime industry, it is crucial to have a staff that is highly skilled and competent [17].

5. FUTURE RESEARCH DIRECTIONS

Given FFF's potential in the marine industry, there are ample opportunities to do considerable research aimed at addressing the current challenges associated with the technology. Scientists are currently investigating the potential for enhancing the ability of materials to resist fouling by including substances like Copper into the filaments [42]. It is worthwhile to investigate other material combinations in order to enhance durability in harsh environmental conditions and increase mechanical qualities. Another potential study field involves addressing challenges related to FFF machines operating in dynamic environmental settings. The utilization of passive dampers in stereolithography (SLA) has demonstrated a significant enhancement in the quality of items manufactured on board [41]. The effectiveness of such strategies can also be investigated for FFF.

Process annealing has demonstrated promising results in enhancing mechanical strength and eliminating the need for post-processing [38]. However, this has only been investigated in specimens used for tensile testing. Additional investigation can be conducted on tall slender wall structures to evaluate the quality of bonding in the z-direction.

The combination of acoustic emission (AE) diagnosis and unsupervised machine learning (ML) has recently allowed for the experimental comparison of a failed FFF printing process with a regular one [51]. The experimental results confirm that the recommended method is capable of diagnosing common process problems, including detection and identifica-

tion. This novel method provides a non-intrusive diagnostic instrument for monitoring the FFF process. Similarly, ML can be employed at the design level to aid in the design for additive manufacturing (DFAM), hence minimizing the likelihood of failure during the initial stages. ML can be advantageous for predictive maintenance. The assessment of aging impacts on the 3D printer's nozzle was conducted utilizing a ML pipeline [23]. This pipeline involved integrating and preparing sensor data, as well as building a regression model. One can investigate the potential application of this approach to other machine components in order to minimize machine downtime. Additionally, there is potential for conducting additional study on the identification of repairable resources using FFF and the optimum parameters for this purpose.

6. CONCLUDING REMARKS

This article examines the potential of FFF in the marine sector. It is clear that for repairing and creating prototypes, it might be seen as a feasible substitute for conventional manufacturing techniques. Nevertheless, the existing limitations greatly restrict its application in direct manufacturing within this area. Specifically, its usage is primarily limited to non-essential equipment and components. Recent research on process enhancement and integration of promising technologies using machine learning has the potential to uncover limitless possibilities. This will enhance its utilization in this industry, leading to a broader adoption.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Rakin Ahmed: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Rounak Saha Niloy:** Conceptualization, Methodology, Investigation, Writing – review & editing. **Md Rahat Mozumder:** Conceptualization, Methodology, Writing – review & editing. **Tanvir Ahmed Shanto:** Conceptualization, Formal analysis, Investigation, Writing – review & editing.

DECLARATION OF COMPETING INTEREST

The authors declare no competing financial interests or personal relationships that could have influenced the work reported.

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